

Benchmark #6

3 MeV H on Fe

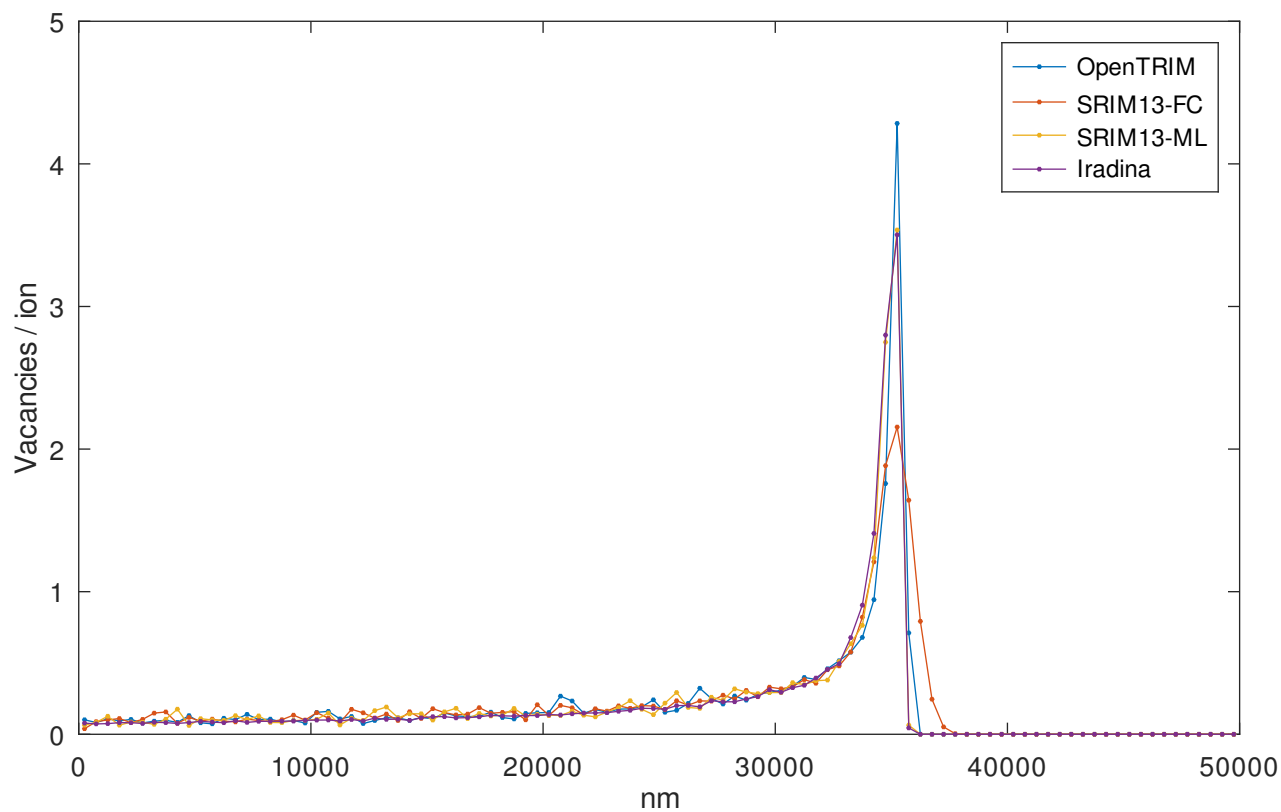
Ion energy $E_0 = 3e+06$ eV

Target depth = 50000 nm

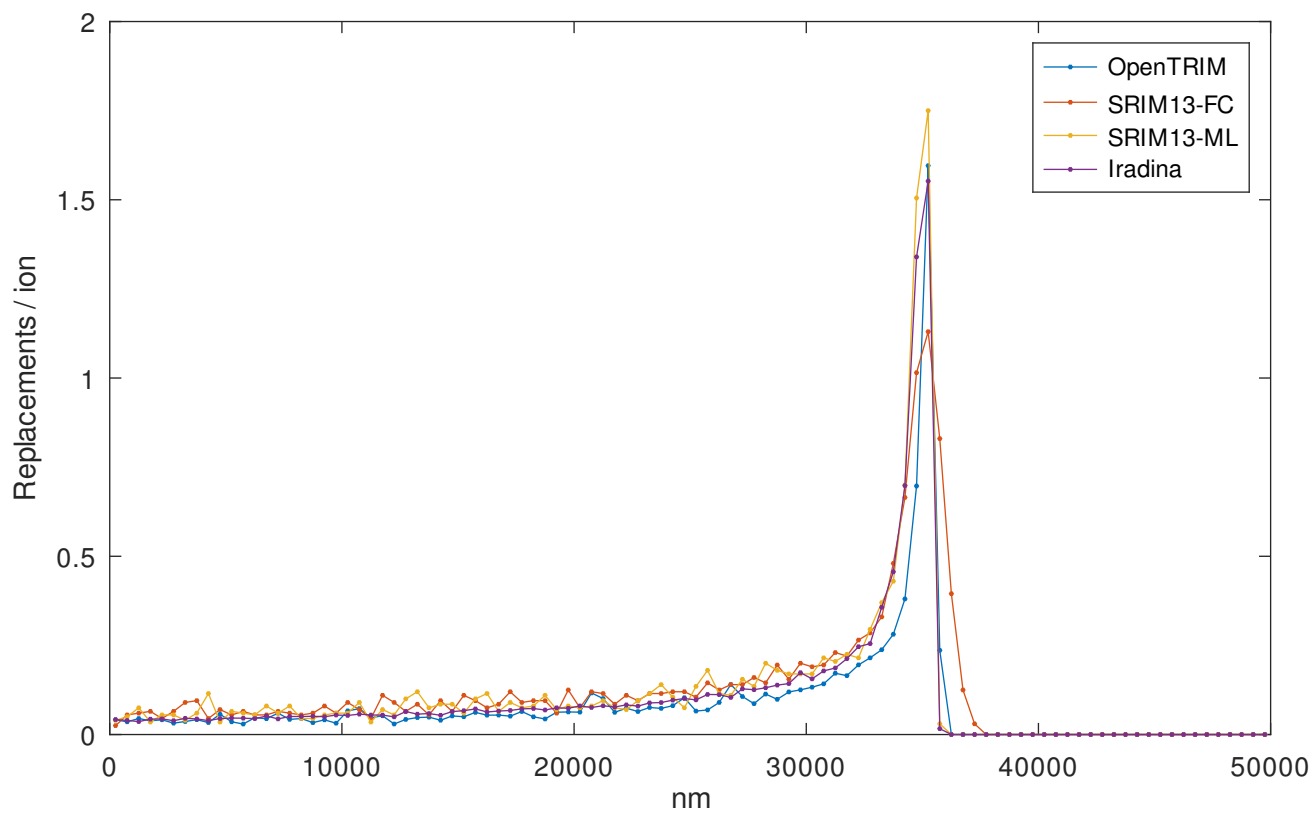
Summary Table

Quantity	OpenTRIM	SRIM13-FC	SRIM13-ML	Iradina
$V(Fe)$	20.1	20.9	20	19.5
$R(tot)$	8.15	11.9	11.4	9.95
$I(H)$	1	1	1	1
$\Delta E_e(H)/E_0$	0.999	0.999	0.999	0
$\Delta E_e(r)/E_0$	6.51e-05	7.53e-05	7.14e-05	0
$\Delta E_e(tot)/E_0$	0.999	0.999	0.999	0.999
$\Delta E_n(H)/E_0$	0.000374	0.000316	0.000486	0
$\Delta E_n(r)/E_0$	0.000558	0.000848	0.000994	0
$\Delta E_n(tot)/E_0$	0.000932	0.00116	0.00148	0.000976
$1 - (\Delta E_e + \Delta E_n)/E_0$	1.01e-05	-0.000302	-0.000382	-4.77e-07

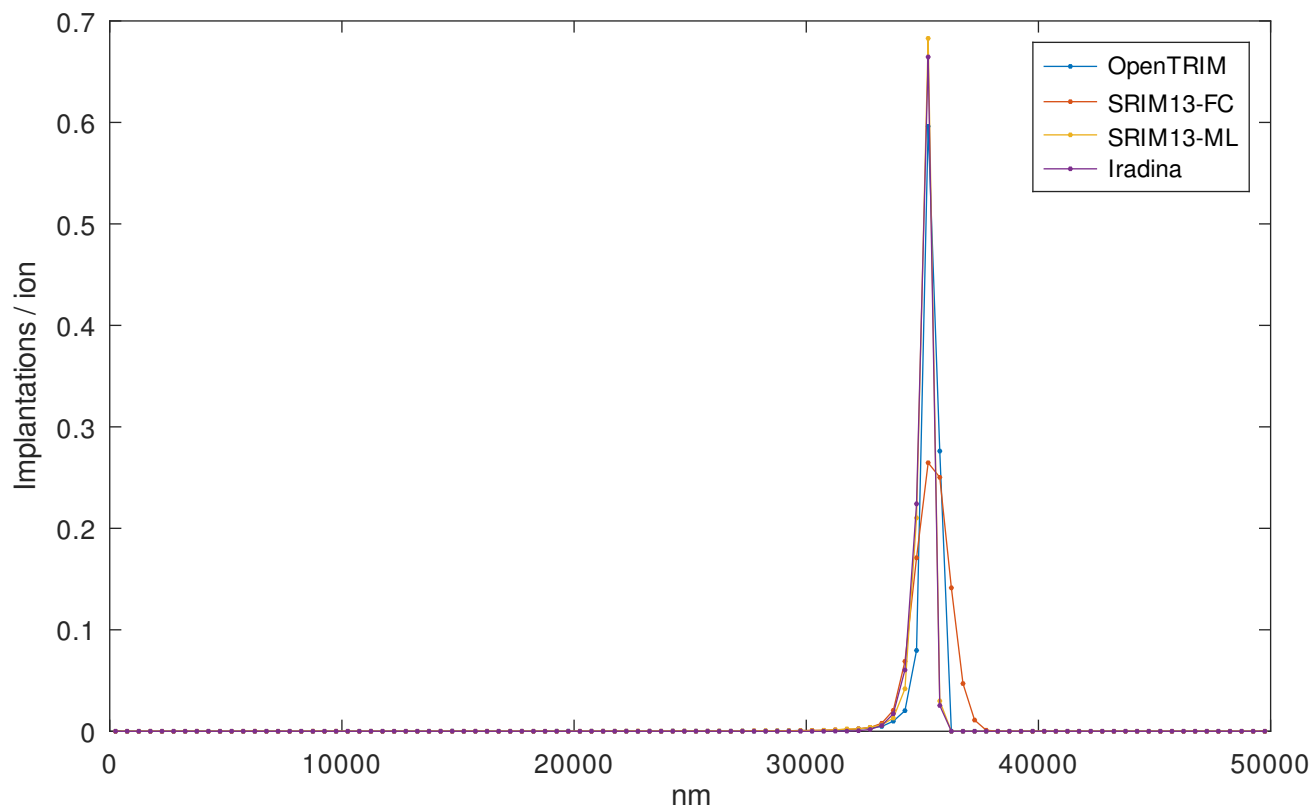
Vacancies of Fe in Fe



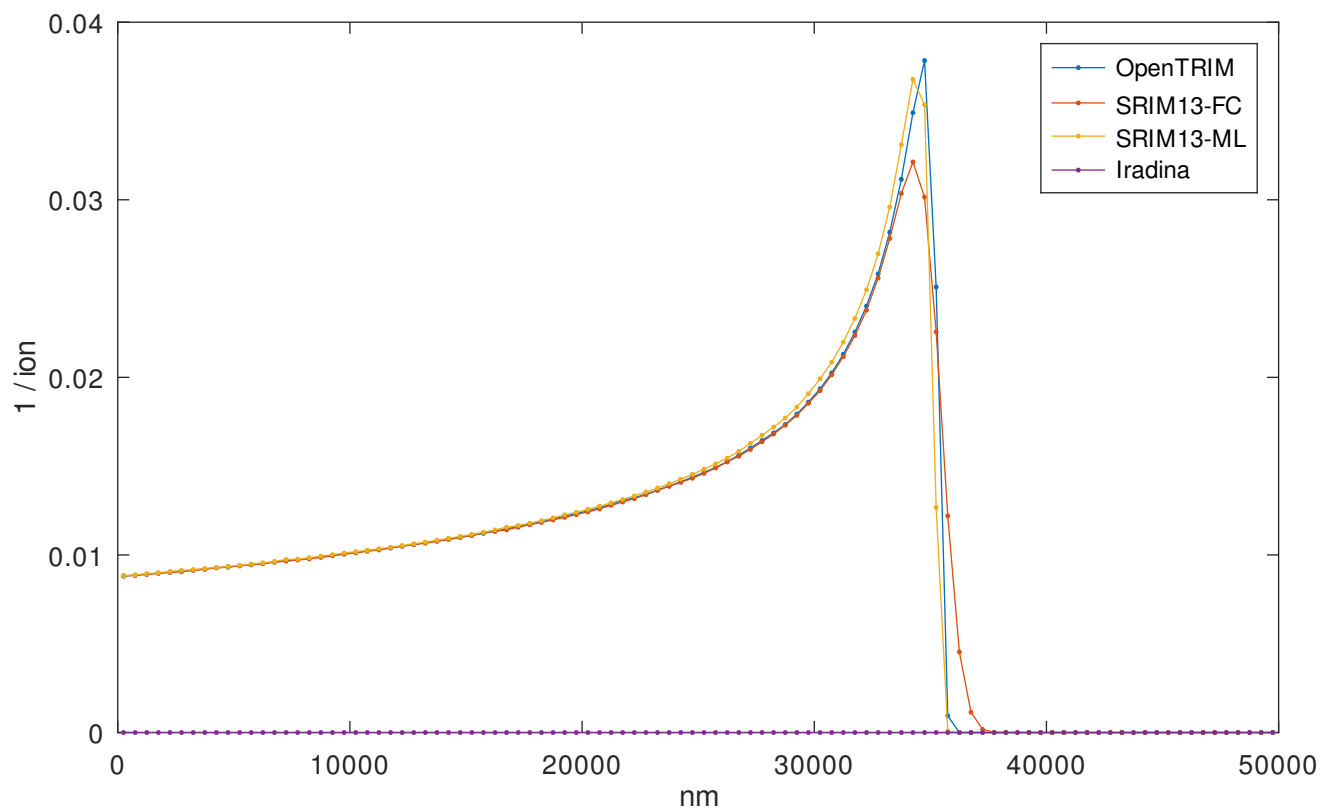
Replacements



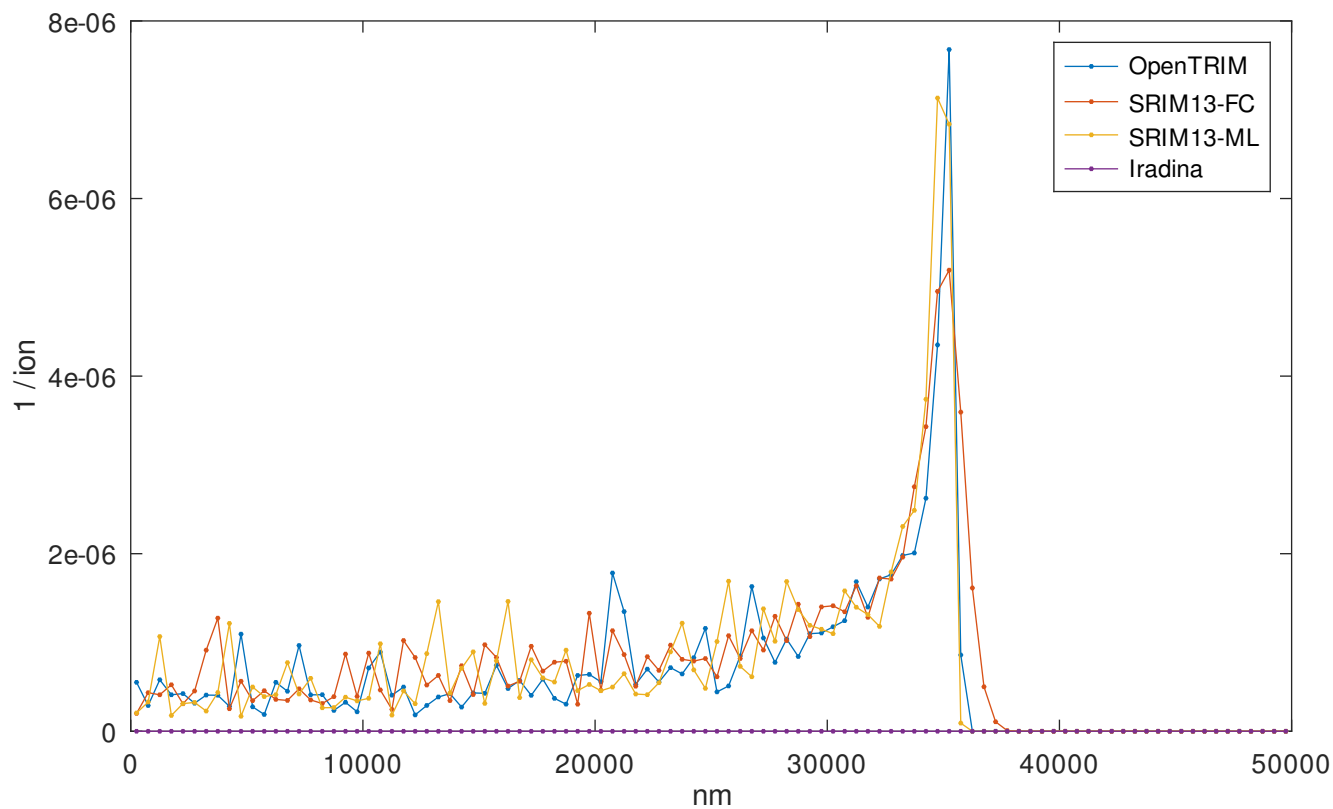
Implanted H ion



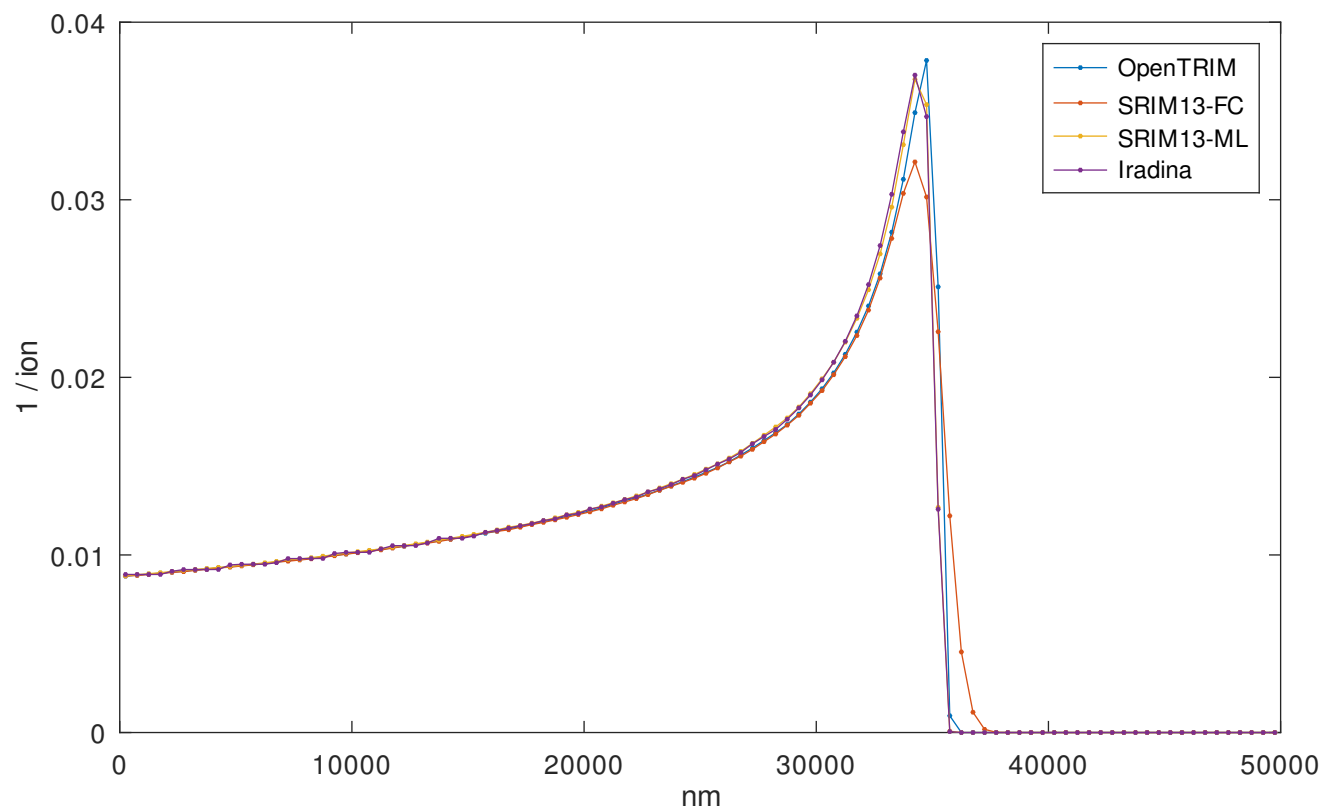
Electronic energy loss fraction $\Delta E_e/E_0$ by H ion



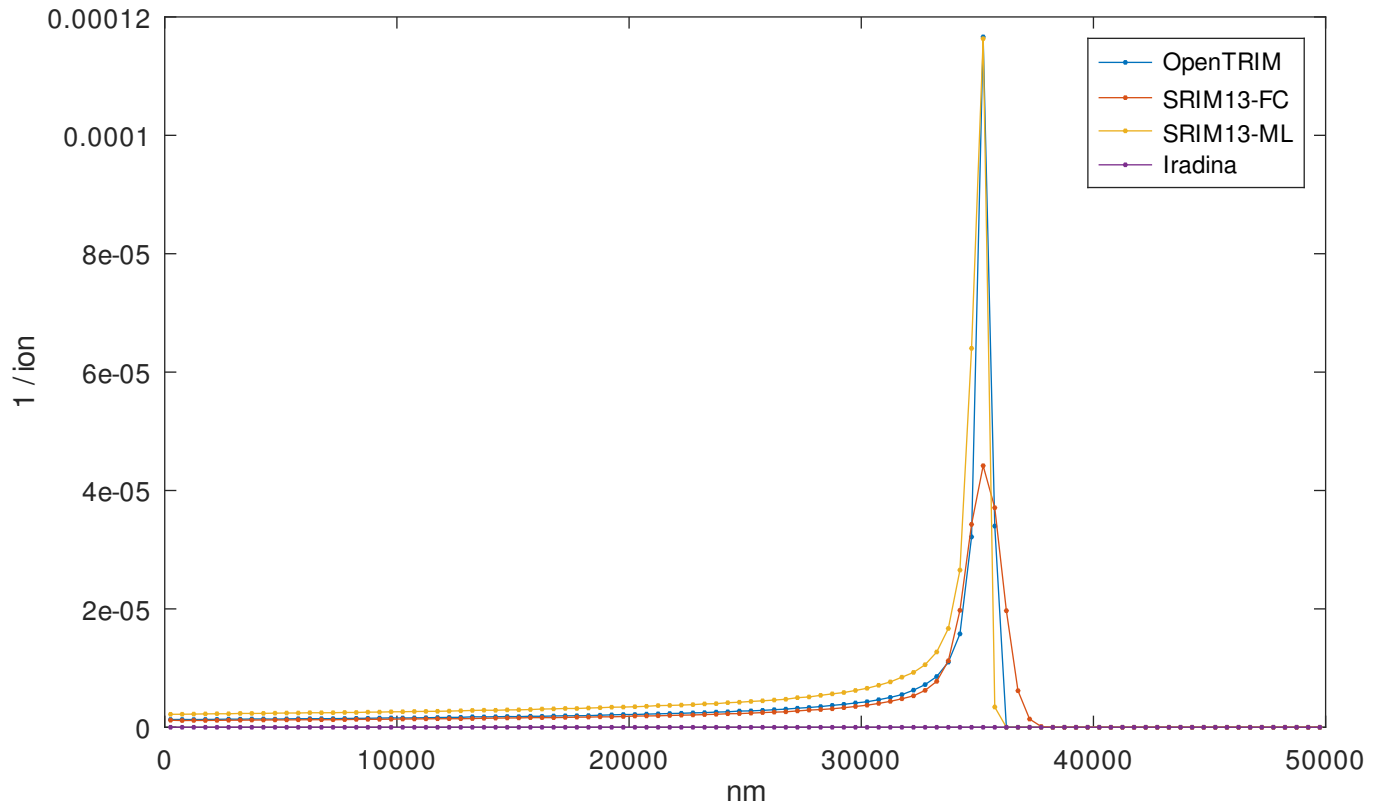
Electronic energy loss fraction $\Delta E_e/E_0$ by recoils



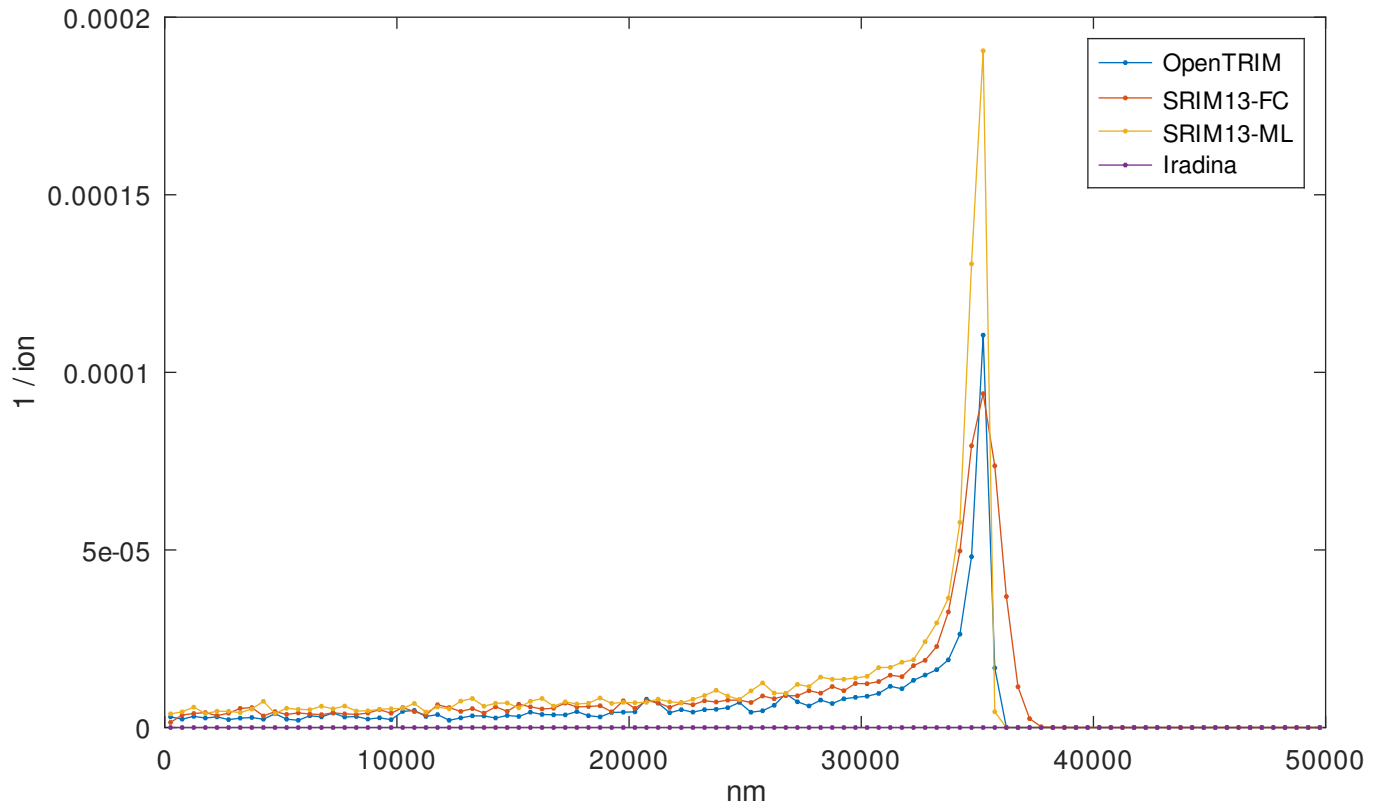
Total Electronic energy loss fraction $\Delta E_e/E_0$



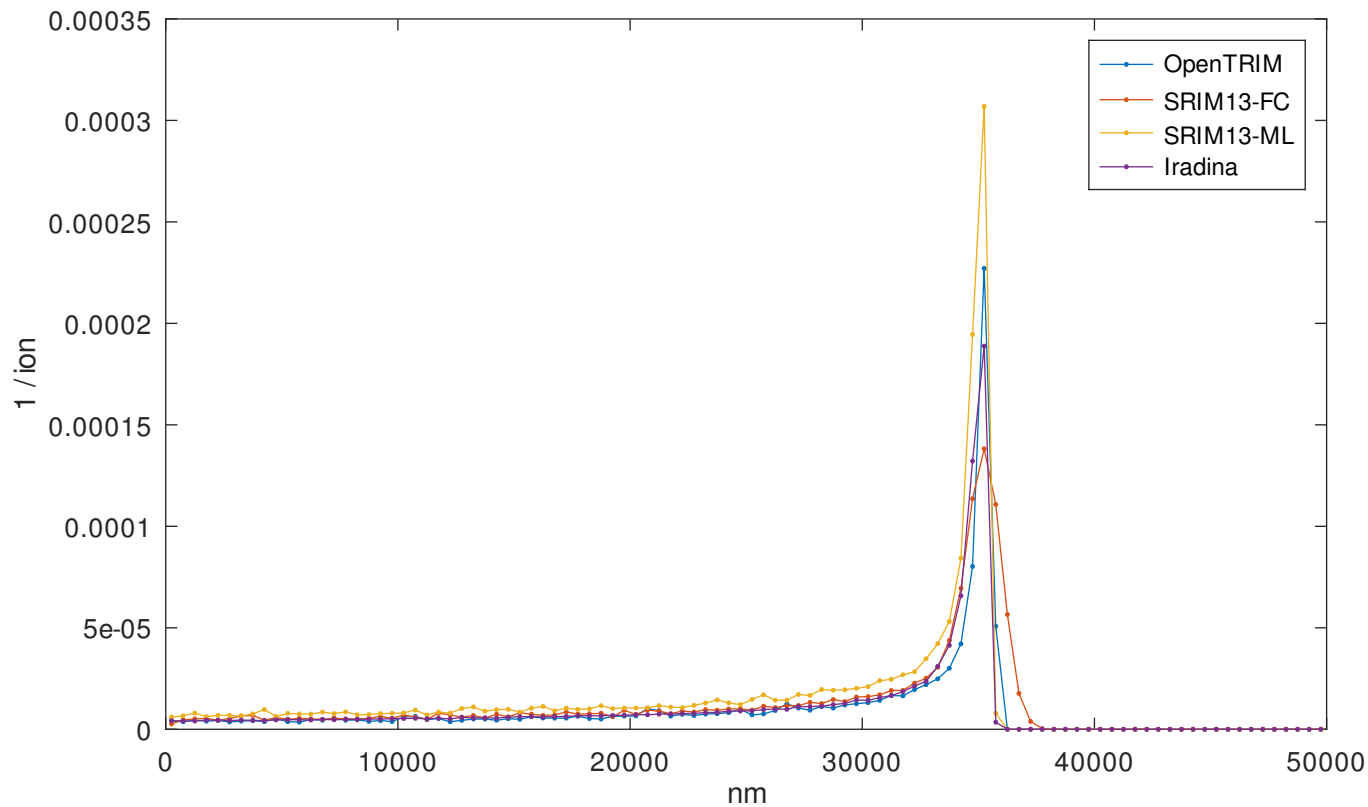
Nuclear energy loss fraction $\Delta E_n/E_0$ by H ion



Nuclear energy loss fraction $\Delta E_n/E_0$ by recoils



Total Nuclear energy loss fraction $\Delta E_n/E_0$



Total fractional energy loss $(\Delta E_e + \Delta E_n)/E_0$

